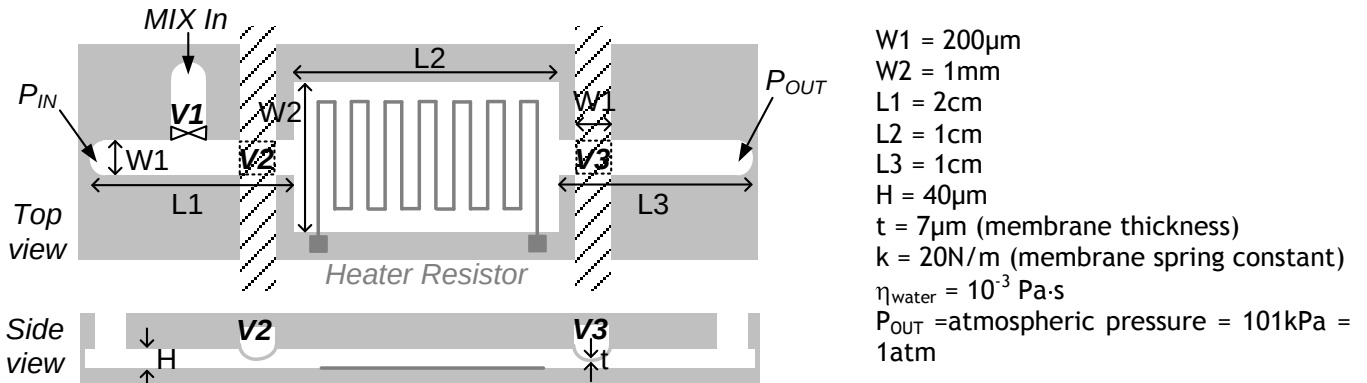


Politecnico di Milano
Biochip A.A. 2013/2014 - M. Carminati
 Appello 3.09.2014

*Test duration: 2 hours - No support material (textbooks, laptops, smartphones, notes) admitted.
 All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.*

Exercise 1

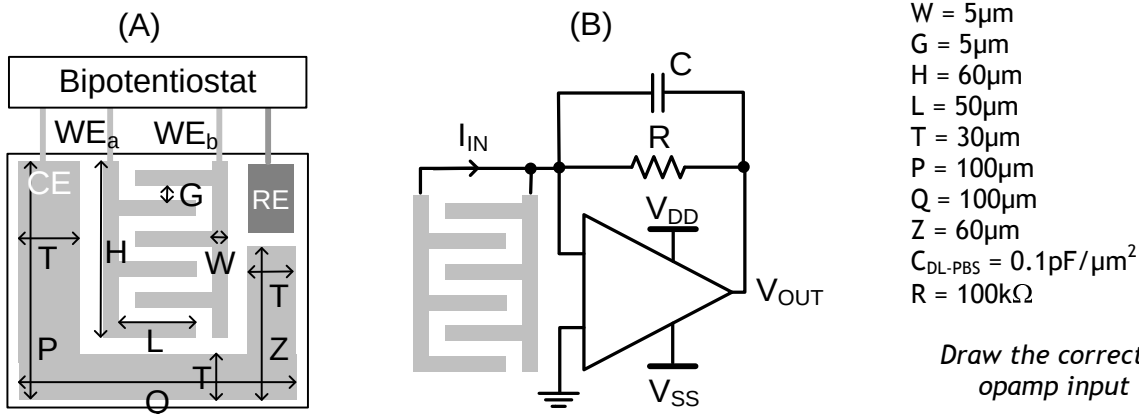
Consider the following PDMS microfluidic system for PCR amplification of the input sample in a single chamber whose temperature is controlled by means of a thin-film resistor. Flow is controlled by 3 valves V1, V2 and V3:



- Given the input pressure $P_{\text{IN}} = 110\text{kPa}$ (generated by an external syringe pump), V1 closed, V2 open and V3 open, and neglecting transients, find the flow speed in the three sections, L1, L2 and L3.
- Briefly describe the operating principles of valves V2 and V3.
- Neglecting the MIX, what is the volume of sample that can be processed and estimate the chamber filling time.
- Considering only V3, find the *hydrodynamic capacitance* associated with the thin square flexible membrane.
- Determine the highest temperature that the liquid in the reaction chamber must reach during the PCR process.
- Draw the temperature temporal sequence $T(\text{time})$ for a complete PCR cycle, describing with a single-line sentence the different phases.
- Propose a system for mixing *on chip* the sample solution and the MIX reagents injected from a separate input.

Exercise 2

Consider the following set of microfabricated planar electrodes, including a pair of interdigitated working electrodes (WE_a and WE_b) with 3 fingers each:



- Draw the detailed scheme of a *Bipotentiostat* connected to the electrodes shown in (A).
- For the *Bipotentiostat* of point a), set the biases in the circuit in DC in order to have the potential of $\text{WE}_a = +0.5\text{V}$ (with respect to the solution), and the potential of $\text{WE}_b = -0.3\text{V}$ (with respect to the solution).
 Now consider both working electrodes connected together to a single current amplifier as shown in (B):
- Determine the value of the double layer capacitance C_{DL} connected at the amplifier input when the electrodes are in PBS. Is the C_{DL} of the counter electrode (CE) negligible?
- Considering an ideal opamp and only the thermal noise of the feedback resistor R, determine the value of the bandwidth BW (and consequently of C) in order to have an input current resolution of 100pA RMS .
- Now considering the input equivalent voltage noise source of the opamp $e_n = 3\text{nV}/\sqrt{\text{Hz}}$, determine the value the corner frequency and plot the input-referred noise spectrum.
- Considering a cyclic voltammetry measure in the macroscopic regime, if the area of the working electrodes is reduced by a factor 2, how should the *scan rate* be modified to measure the same input current?
- What is the spatial accuracy of the *mask aligner* necessary to fabricate the electrodes in (A)?
- Propose a scheme to measure the total current flowing from the COUNTER electrode (CE).